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系所: CS 2B

1. Describe an $O(n)$ -time algorithm that, given a set S of n distinct numbers and a positive integer $k \leq n$, finds the k elements in S whose values are closest to the median of S (excluding the median itself).

Hint: The selected elements need not be symmetrically distributed around the median in the sorted order. For example, given 1, 2, 3, 8, 10, the two elements closest to the median 3 are 1 and 2 (excluding the median), both of which lie on the same side. This is only an illustration; do not assume the input is sorted.

ClosestMedian(S, k) $m \leftarrow \text{SELECTS}(S, \frac{n}{2})$ create an array D for $i \leftarrow 1 \sim n$ $D[i] \leftarrow \text{abs}(S[i] - m)$ $d \leftarrow \text{SELECTS}(D, k+1)$ create an array ans for $i \leftarrow 1 \sim n$ if $(\text{abs}(S[i] - m) \leq d)$ and $(\text{abs}(S[i] - m) \neq 0)$:push $S[i]$ to ans return ans

) 1. Use linear Select Algorithm to find set's median.

) 2. 計算每個數離 median 的距離

) 3. 找第 $k+1$ 小的距離 d) 4. 找差值 $\leq d$ 且不是中位數本身的數你代代看 $[1, 2, 3], k=1$

$$1. + 2. + 3. + 4. = O(n) + O(n) + O(n) + O(n) = O(n)$$